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NO.1 PVT. UNIVERSITY IN
ACADEMIC REPUTATION IN INDIA



ACCREDITED **GRADE 'A'**
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PERFECT SCORE OF **150/150** AS A TESTAMENT
TO EXCEPTIONAL E-LEARNING METHODS

Unit 2 : Logic for AI

Prepared by

School of Computer Science
UPES, Dehradun
India

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Introduction to Logic

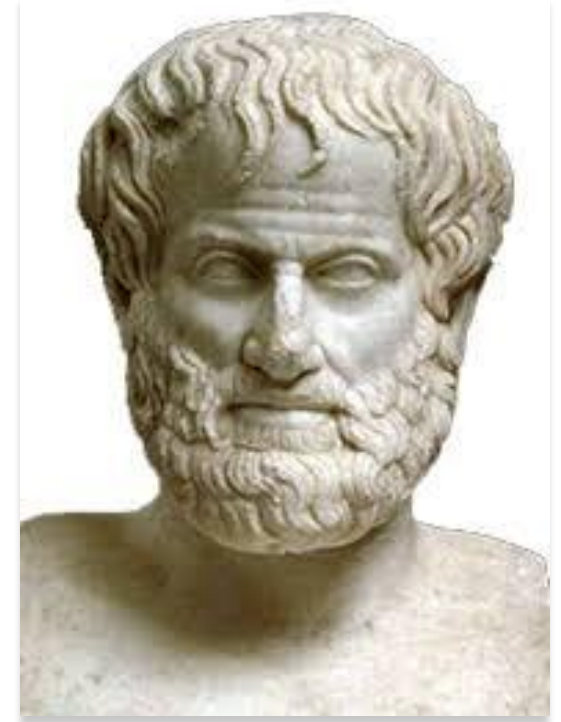
What is Logic?

The study of principles and methods used to distinguish **correct** reasoning from **incorrect** reasoning.

Logic is a formal way of representing knowledge using facts, rules, and relationships, and then reasoning from them to reach conclusions.

Father of Logic

Aristotle (384–322 BCE): Often considered the "father of logic," **Aristotle** developed the first formal system of logic, known as **syllogistic logic**. His work, particularly in "Organon," laid the groundwork for future logical studies.



Propositional Logic

- Propositional logic is also known as **propositional calculus** or **sentential logic**.
- It is a branch of logic that deals with **propositions** and their relationships through logical **connectives**.
- It forms the foundation for many areas of formal logic, computer science, and mathematics.

Propositions

- A proposition is a declarative statement that is either **True** or **False** but not both.
- Examples include:
 - "It is raining" (True or False)
 - " $2 + 2 = 4$ " (True).

Logical Connectives

Connectives: Symbols representing logical operations.

Types of Connectives:

- *Negation* ($\neg$) : The negation of a proposition P is denoted as $\neg P$ and is True if P is False, and False if P is True.
- *Conjunction* ($\wedge$) : The conjunction of propositions P and Q is denoted as $P \wedge Q$ and is True if both P and Q are True.
- *Disjunction* ($\vee$) : The disjunction of propositions P and Q is denoted as $P \vee Q$ and is True any one of P or Q is True.
- *Implication* ($\rightarrow$) : The implication $P \rightarrow Q$ is true if whenever P is True, Q is also True. It is False only if P is True and Q is False.
- *Bi-implication* ($\leftrightarrow$) : The biconditional $P \leftrightarrow Q$ is true if P and Q are either both True or both False.

Truth Tables

Purpose: Systematically determine the truth value of compound propositions.

Construction: Assign truth values to variables and apply logical connectives.

Example: Truth table for $(p \wedge q) \vee \neg p$

Truth Table Examples

P	$\neg P$
T	F
F	T

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

P	Q	$P \leftrightarrow Q$
T	T	T
T	F	F
F	T	F
F	F	T

For a detailed tutorial on Truth Tables, please refer to:
<https://sites.millersville.edu/bikenaga/math-proof/truth-tables/truth-tables.html>

Laws of Propositional Logic

- **Commutative Laws:** $p \wedge q \equiv q \wedge p$
- **Associative Laws:** $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
- **Distributive Laws:** $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
- **De Morgan's Laws:** $\neg(p \wedge q) \equiv \neg p \vee \neg q$
- **Idempotent Laws:** $p \wedge p \equiv p$

Example: Applying De Morgan's Law: $\neg(p \vee q) \equiv \neg p \wedge \neg q$

Examples

- Simplify the expression $\neg(p \vee q) \wedge (p \rightarrow q)$

Solution

Simplifying the expression $\neg(p \vee q) \wedge (p \rightarrow q)$

1. Apply De Morgan's Law to $\neg(p \vee q)$:

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

2. Apply the definition of implication to $p \rightarrow q$:

$$p \rightarrow q \equiv \neg p \vee q$$

3. Substitute these into the original expression:

$$(\neg p \wedge \neg q) \wedge (\neg p \vee q)$$

4. Distribute $\neg p$ across $(\neg p \wedge \neg q)$ and $(\neg p \vee q)$:

$$(\neg p \wedge \neg q) \wedge (\neg p \vee q) \equiv (\neg p \wedge \neg q \wedge \neg p) \vee (\neg p \wedge \neg q \wedge q)$$

5. Simplify the expression:

$$(\neg p \wedge \neg q \wedge \neg p) \vee (\neg p \wedge \neg q \wedge q) \equiv \neg p \wedge \neg q$$

So, the simplified expression is:

$$\neg p \wedge \neg q$$

Terminology for implication.

Because implication statements play such an essential role in mathematics, a variety of terminology is used to express $p \rightarrow q$:

- “if p , then q ”.
- “ q , if p ”.
- “ p , only if q ”.
- “ p implies q ”.
- “ p is sufficient for q ”.
- “ q is necessary for p ”.
- “ q follows from p ”.

Exclusive Or operator

Truth table for Exclusive Or $\oplus$

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

- Actually, this operator can be expressed by using other operators:
 $p \oplus q$ is the same as $\neg (p \leftrightarrow q)$.
- $\oplus$ is used often in CSE. So we have a symbol for it.

Precedence of Operators

Operator	Precedence
$\neg$	1
$\wedge$	2
$\vee$	3
$\rightarrow$	4
$\leftrightarrow$	5

Example:

$\neg p \wedge q$ means $(\neg p) \wedge q$
 $p \wedge q \rightarrow r$ means $(p \wedge q) \rightarrow r$

- When in doubt, use parenthesis.

Translating logical formulas to English sentences

Using the above logic operators, we can construct more complicated logical formulas. (They are called **compound propositions**.)

Example

Proposition p : Alice is smart.

Proposition q : Alice is honest.

$\neg p \wedge q$: Alice is not smart but honest.

$p \vee (\neg p \wedge q)$: Either Alice is smart, or she is not smart but honest.

$p \rightarrow \neg q$: If Alice is smart, then she is not honest.

Translating logical formulas from English sentences

We can also go in the other direction, translating English sentences to logical formulas:

- Alice is either smart or honest, but Alice is not honest if she is smart:

$$(p \vee q) \wedge (p \rightarrow \neg q).$$

- That Alice is smart is necessary and sufficient for Alice to be honest:

$$(p \rightarrow q) \wedge (q \rightarrow p).$$

(This is often written as $p \leftrightarrow q$).

Examples of Logical equivalence

Example:

Look at the following two compound propositions: $p \rightarrow q$ and $q \vee \neg p$.

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

p	q	$\neg p$	$q \vee \neg p$
T	T	F	T
T	F	F	F
F	T	T	T
F	F	T	T

- The last column of the two truth tables are identical. Therefore $(p \rightarrow q)$ and $(q \vee \neg p)$ are **logically equivalent**.
- So $(p \rightarrow q) \leftrightarrow (q \vee \neg p)$ is a tautology.
- Thus: $(p \rightarrow q) \equiv (q \vee \neg p)$.

Example:

By using truth table, prove $p \oplus q \equiv \neg (p \leftrightarrow q)$.

De Morgan law

We have a number of rules for logical equivalence. For example:

De Morgan Law:

$$\neg(p \wedge q) \equiv \neg p \vee \neg q \quad (1)$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q \quad (2)$$

The following is the truth table proof for (1). The proof for (2) is similar.

p	q	$p \wedge q$	$\neg(p \wedge q)$
T	T	T	F
T	F	F	T
F	T	F	T
F	F	F	T

p	q	$\neg p \vee \neg q$
T	T	F
T	F	T
F	T	T
F	F	T

Distributivity

Distributivity

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r) \quad (1)$$

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r) \quad (2)$$

The following is the truth table proof of (1). The proof of (2) is similar.

p	q	r	$q \wedge r$	$p \vee (q \wedge r)$	$p \vee q$	$p \vee r$	$(p \vee q) \wedge (p \vee r)$
T	T	T	T	T	T	T	T
T	T	F	F	T	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	T	T	T	T
F	T	T	T	T	T	T	T
F	T	F	F	F	T	F	F
F	F	T	F	F	F	T	F
F	F	F	F	F	F	F	F

Logic Equivalences

Equivalence	Name
$p \wedge T \equiv p, \quad p \vee F \equiv p$	Identity laws
$p \vee T \equiv T, \quad p \wedge F \equiv F$	Domination laws
$p \vee p \equiv p, \quad p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws
$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \vee q) \equiv \neg p \wedge \neg q$ $\neg(p \wedge q) \equiv \neg p \vee \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv T, \quad p \wedge \neg p \equiv F$	Negation laws

Logic Equivalences

Logical Equivalences Involving Conditional Statements

$p \rightarrow q \equiv \neg p \vee q$
$p \rightarrow q \equiv \neg q \rightarrow \neg p$
$p \vee q \equiv \neg p \rightarrow q$
$p \wedge q \equiv \neg(p \rightarrow \neg q)$
$\neg(p \rightarrow q) \equiv p \wedge \neg q$
$(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$
$(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$
$(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$
$(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$

Logical Equivalences Involving Biconditional Statements

$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$
$p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$
$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$
$\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$

Prove equivalence

By using these laws, we can prove two propositions are logical equivalent.

Example 1: Prove $\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg q$.

$$\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg(\neg p \wedge q)$$

DeMorgan

$$\equiv \neg p \wedge (p \vee \neg q)$$

DeMorgan

$$\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q)$$

Distributivity

$$\equiv \mathbf{F} \vee (\neg p \wedge \neg q)$$

Because $\neg p \wedge p \equiv \mathbf{F}$

$$\equiv \neg p \wedge \neg q$$

Because $\mathbf{F} \vee r \equiv r$ for any r

Prove equivalence

Example 2: Prove $(p \wedge q) \rightarrow (p \vee q) \equiv \text{T}$.

$$\begin{aligned}(p \wedge q) \rightarrow (p \vee q) &\equiv \neg(p \wedge q) \vee (p \vee q) && \text{Substitution for } \rightarrow \\ &\equiv (\neg p \vee \neg q) \vee (p \vee q) && \text{DeMorgan} \\ &\equiv (\neg p \vee p) \vee (\neg q \vee q) && \text{Commutativity and Associativity} \\ &\equiv \text{T} \vee \text{T} && \text{Because } \neg p \vee p \equiv \text{T} \\ &\equiv \text{T}\end{aligned}$$

Prove equivalence

Example 3:

Prove $(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$.

Note that, by “Substitution for $\rightarrow$ ”, we have: $\text{RHS} = \neg p \vee (q \wedge r)$.
So, we start from the LHS and try to get this proposition:

$(p \rightarrow q) \wedge (p \rightarrow r) \equiv (\neg p \vee q) \wedge (\neg p \vee r)$	Substitution for $\rightarrow$, twice
$\equiv \neg p \vee (q \wedge r)$	Distribution law
$\equiv p \rightarrow (q \wedge r)$	Substitution for $\rightarrow$

What is Predicate Logic?

Predicate Logic: Deals with predicates and quantified variables.

Predicates: Statements containing variables that can be either true or false.

Quantifiers: Symbols indicating the extent of the variable's applicability (e.g., $\forall$ for universal quantification, $\exists$ for existential quantification).

Example: $\forall x(P(x))$ represents "for all x , $P(x)$ ".

Predicates and Quantifiers

Predicates: Expressions containing variables that can be true or false.

Quantifiers: Determine the scope of variables.

Examples:

Predicate: $P(x)$ - "x is a prime number."

Quantified Statement: $\forall x(P(x))$ - "All x are prime numbers."

Difference between Propositional and Predicate Logic

→ **Propositional logic** deals with entire statements as atomic units.

- A statement is either: True (T) or False (F)
- It does not analyze internal structure of the statement.

Example

Let:

P : "It is raining"

Q : "The road is wet"

Formula: $P \rightarrow Q$

Meaning: If it is raining, then the road is wet.

Difference between Propositional and Predicate Logic

→ **Predicate Logic (First-Order Logic – FOL)** analyzes the **internal structure** of statements.

- It introduces: Objects, Properties, Relations, and Quantifiers

Syntax Components

Predicates: $Rain(x)$, $Wet(x)$

Variables: x, y

Quantifiers:

$\forall$ (for all)

$\exists$ (there exists)

Example

Instead of: $P \rightarrow Q$

We write: $\forall x(Rain(x) \rightarrow Wet(x))$

Meaning: For every object x , if it is raining at x , then x is wet.

Difference between Propositional and Predicate Logic

Feature	Propositional Logic	Predicate Logic
Basic Unit	Whole statement	Predicate with variables
Variables	No object variables	Uses variables (x, y)
Quantifiers	Not allowed	$\forall, \exists$ allowed
Expressiveness	Limited	Highly expressive
Example	$P \rightarrow Q$	$\forall x (\text{Student}(x) \rightarrow \text{Passed}(x))$
Internal Structure	Ignored	Analyzed

Difference between Propositional and Predicate Logic

Example:

Statement: “All students in this class passed the exam.”

In Propositional Logic

You must write:

$$P_1 \wedge P_2 \wedge P_3 \wedge \dots$$

If new students join $\rightarrow$ rewrite everything.

In Predicate Logic

$$\forall x (Student(x) \rightarrow Passed(x))$$

Quantifier Negation

Negating Quantified Statements:

Universal Quantification: $\neg(\forall x(P(x))) \equiv \exists x(\neg P(x))$

Existential Quantification: $\neg(\exists x(P(x))) \equiv \forall x(\neg P(x))$

Example:

Negating "All cats are mammals"
becomes
"Some cats are not mammals".

Rules of Inference

- **Modus Ponens:** If p implies q and p is true, then q is true.
- **Modus Tollens:** If p implies q and q is false, then p is false.
- **Hypothetical Syllogism:** If p implies q , and q implies r , then p implies r .
- **Disjunctive Syllogism:** If p or q is true, and $\neg p$ is true, then q is true.
- **Constructive Dilemma:** If p implies q , and r implies s , and $p \vee r$ is true, then $q \vee s$ is true.

Example: Using Modus Ponens: If "It is raining" implies "The ground is wet", and "It is raining", then "The ground is wet".

Literal Meaning:

- **Modus** = *mode* or *method*
 - **Ponens** = *affirming* (from *ponere*, meaning "to place" or "to assert")
- So, Modus Ponens literally means:
"The method of affirming."

Modus Tollens → denies (rejects) the consequent

Syllogism: "a logical inference" or "a conclusion drawn from premises."

1. Modus Ponens (Law of Detachment)

If a conditional statement ("if-then" statement) is true, and its antecedent (the "if" part) is true, then its consequent (the "then" part) must also be true.

Form: If $p \rightarrow q$ and p , then q .

Example:

- *Premise: If it rains, the ground will be wet.*
- *Premise: It is raining.*
- *Conclusion: The ground is wet.*

2. Modus Tollens (Law of Contrapositive)

If a conditional statement is true, and its consequent is false, then its antecedent must also be false.

Form: If $p \rightarrow q$ and $\neg q$, then $\neg p$.

Example:

- *Premise: If it rains, the ground will be wet.*
- *Premise: The ground is not wet.*
- *Conclusion: It is not raining.*

3. Hypothetical Syllogism

If two conditional statements are true, where the consequent of the first is the antecedent of the second, then a third conditional statement combining the antecedent of the first and the consequent of the second is also true.

Form: *If $p \rightarrow q$ and $q \rightarrow r$, then $p \rightarrow r$.*

Example:

- *Premise: If it rains, the ground will be wet.*
- *Premise: If the ground is wet, the plants will grow.*
- *Conclusion: If it rains, the plants will grow.*

4. Disjunctive Syllogism

If a disjunction (an "or" statement) is true, and one of the disjuncts (the parts of the "or" statement) is false, then the other disjunct must be true.

Form: *If $p \vee q$ and $\neg p$, then q .*

Example:

- *Premise: It is either raining or sunny.*
- *Premise: It is not raining.*
- *Conclusion: It is sunny.*

CONSTRUCTIVE DILEMMA

1. $P \Rightarrow Q$

2. $R \Rightarrow S$

3. $P \vee R$

$\therefore Q \vee S$

1. IF I AM RUNNING, I AM HAPPY.

2. IF I AM SLEEPING, I AM DREAMING.

3. I AM RUNNING OR SLEEPING.

$\therefore$ THEREFORE, I AM HAPPY OR I AM DREAMING.

5. Conjunction

If two statements are true, then their conjunction (an "and" statement) is also true.

Form: If p and q , then $p \wedge q$.

Example:

- *Premise: It is raining.*
- *Premise: It is windy.*
- *Conclusion: It is raining and windy.*

6. Simplification

If a conjunction (an "and" statement) is true, then each of its conjuncts is also true.

Form: If $p \wedge q$, then p

Example:

- *Premise: It is raining and windy.*
- *Conclusion: It is raining.*

7. Addition

If a statement is true, then the disjunction (an "or" statement) of that statement with any other statement is also true.

Form: If p , then $p \vee q$

Example:

- *Premise: It is raining.*
- *Conclusion: It is raining or sunny.*

Predicate Logic Inference Rules

- **Universal Instantiation:** If $\forall x(P(x))$ is true, then $P(a)$ is true for any individual a .
- **Existential Instantiation:** If $\exists x(P(x))$ is true, then $P(a)$ is true for some particular a .
- **Universal Generalization:** If $P(a)$ is true for any individual a , then $\forall x(P(x))$ is true.
- **Existential Generalization:** If $P(a)$ is true for some particular a , then $\exists x(P(x))$ is true.

Example: Using Universal Instantiation: If "All cats are mammals", then "Fluffy is a mammal".

Logical Equivalences involving Predicates and Quantifiers

Definition: Two logical statements involving predicates and quantifiers are said to be equivalent if and only if they have the same truth value in all the possible cases.

Logical equivalence is helpful in replacing one expression with an equivalent expression.

There are two important equivalences involving quantifiers:

$$\forall x (P(x) \wedge Q(x)) \equiv \forall x P(x) \wedge \forall x Q(x)$$

$$\exists x (P(x) \vee Q(x)) \equiv \exists x P(x) \vee \exists x Q(x)$$

Note: we can also use $\Leftrightarrow$ symbol instead of $\equiv$ symbol.

Predicate Logic Equivalences

- **Universal Quantifier Equivalence:**

$$\neg \forall x(P(x)) \equiv \exists x(\neg P(x))$$

- **Existential Quantifier Equivalence:**

$$\neg \exists x(P(x)) \equiv \forall x(\neg P(x))$$

- **Negation Equivalence:**

$$\neg(\forall x(P(x))) \equiv \exists x(\neg P(x)) \text{ and } \neg(\exists x(P(x))) \equiv \forall x(\neg P(x))$$

Consider the following statement:

"Every student in the xyz university has studied discrete mathematics."

Domain: All students of xyz university.

$P(x)$: x has studied discrete mathematics.

"Every student in the xyz university has studied discrete mathematics."

is equivalent to

$$\forall x P(x)$$

What is the negation of the above statement?

Consider the following statement:

"Every student in the xyz university has studied discrete mathematics."

Domain: All students of xyz university.

$P(x)$: x has studied discrete mathematics.

"Every student in the xyz university has studied discrete mathematics."

is equivalent to

$$\forall x P(x)$$

What is the negation of the above statement?

"It is not the case that every student in the xyz university has studied discrete mathematics."

Applications of Logic

Mathematics: Proving theorems, solving equations.

Computer Science: Designing algorithms, programming languages.

Philosophy: Analyzing arguments, studying language.

Linguistics: Understanding sentence structure, semantics.

Example: Using logic to verify software correctness.

Propositional Logic	Predicate Logic
The values of this logic can be either true or false but they cannot be ambiguous.	The values in a predicate table depend on the true and false values of different variables.
Logical operators are used in propositional logic	Predicate logic is applied by adding quantifiers to a proposition.
It is a general logic	It is a special logic.
This is a logic in which different types of declarative statements are included whose values can be true or false.	This is a logic which consists of objects and functions and the relation between them.
Scope analysis is not part of propositional logic.	The scope of the subject over the predicate is analyzed in this logic. Three quantifiers are used which include ? ■ Universal Quantifier ■ Existential Quantifier ■ Uniqueness Quantifier
It cannot be used to deal with a set of entities.	Quantifiers are used in this logic to deal with a set of entities.
Propositional logic is also known as Boolean logic as the result can be either true or false. This is the reason it is called basic logic.	This is an advanced level of propositional logic in which the main points that are to be covered which include predicates and quantifications.

Uniqueness Quantifiers

→ The **uniqueness quantifier** expresses that **there exists exactly one element** in the domain that satisfies a given property.

→ **Symbol:** The uniqueness quantifier is written as: $\exists!$

Read as:

“There exists exactly one”

→ **Basic Form**

$$\exists! x P(x)$$

Meaning: There exists exactly one x such that $P(x)$ is true.

Uniqueness Quantifiers

Formal Definition:

The uniqueness quantifier is defined using ordinary existential and universal quantifiers.

$$\exists! x P(x)$$

is logically equivalent to: $\exists x(P(x) \wedge \forall y(P(y) \rightarrow y = x))$

It has **two parts**:

1.Existence part

$$\exists x P(x)$$

1. $\rightarrow$ At least one object satisfies $P(x)$.

2.Uniqueness part

$$\forall y(P(y) \rightarrow y = x)$$

1. $\rightarrow$ If any object satisfies P , it must be the same object.

So together:

One object satisfies the property, and no other object does.

Uniqueness Quantifiers

Example -

Statement: There exists exactly one even prime number.

Let:

$$P(x) = \text{"}x \text{ is prime and even"}$$

We write:

$$\exists! x P(x)$$

This is true in $\mathbb{N}$ because only **2** satisfies the property.

Difference Between $\exists$ and $\exists!$

Symbol	Meaning	Example
$\exists xP(x)$	At least one	There exists a student who scored 100
$\exists! xP(x)$	Exactly one	There exists exactly one topper

Key idea:

$\exists! \Rightarrow \exists + \textit{uniqueness condition}$

1. Marcus was a man.
2. Marcus was a Pompeian.
3. All Pompeian were Romans.
4. Caesar was a ruler.
5. All Romans were either loyal to Caesar or hated him.
6. Everyone is loyal to someone.
7. People only try to assassinate rulers they are not loyal to.
8. Marcus tried to assassinate Caesar.
9. All men are people

- | | |
|---|---|
| 1. Marcus was a man. | 1. $man(Marcus)$ |
| 2. Marcus was a Pompeian. | 2. $Pompeian(Marcus)$ |
| 3. All Pompeian were Romans. | 3. $\forall x : Pompeian(x) \rightarrow Roman(x)$ |
| 4. Caesar was a ruler. | 4. $ruler(Caesar)$ |
| 5. All Romans were either loyal to Caesar or hated him. | 5. $\forall x : Roman(x) \rightarrow loyalto(x, Caesar) \vee hate(x, Caesar)$ |
| 6. Everyone is loyal to someone. | 6. $\forall x : \exists y : loyalto(x, y)$ |
| 7. People only try to assassinate rulers they are not loyal to. | 7. $\forall x : \forall y : person(x) \wedge ruler(y) \wedge tryassassinate(x, y) \rightarrow \neg loyalto(x, y)$ |
| 8. Marcus tried to assassinate Caesar. | 8. $tryassassinate(Marcus, Caesar)$ |
| 9. All men are people | 9. $\forall x : man(x) \rightarrow person(x)$ |

Practice Question

Write the following statements in predicate logic, and then negate them:

- i. "Every student in the class has submitted the assignment. "
- ii. "There exists a book that is not on the reading list."

Solution

Statement: "Every student in the class has submitted the assignment. "

Let:

Student(x): x is a student

InClass(x) : x is in the class

Submitted(x) : x has submitted the assignment

$$\forall x \left((Student(x) \wedge InClass(x)) \rightarrow Submitted(x) \right)$$

Negate the whole statement:

$$\neg \forall x \left((Student(x) \wedge InClass(x)) \rightarrow Submitted(x) \right)$$

$$\exists x \left((Student(x) \wedge InClass(x)) \wedge \neg Submitted(x) \right)$$

Meaning:

There exists a student in the class who has not submitted the assignment.

- Let:

- $Book(x)$: x is a book

- $OnList(x)$: x is on the reading list

$$\exists x (Book(x) \wedge \neg OnList(x))$$

- Negate:

$$\neg \exists x (Book(x) \wedge \neg OnList(x))$$

- Use rule:

$$\neg \exists x P(x) \equiv \forall x \neg P(x)$$

$$\forall x \neg (Book(x) \wedge \neg OnList(x))$$

- Now use De Morgan's Law:

$$\neg (P \wedge Q) \equiv \neg P \vee \neg Q$$

- So:

$$\forall x (\neg Book(x) \vee OnList(x))$$

- Equivalent form:

$$\forall x (Book(x) \rightarrow OnList(x))$$

- Meaning:

Every book is on the reading list.

Practice Question

What is the correct translation of the following statement into mathematical logic? “Some real numbers are rational”

- (A) $\exists x (\text{real}(x) \vee \text{rational}(x))$
- (B) $\forall x (\text{real}(x) \rightarrow \text{rational}(x))$
- (C) $\exists x (\text{real}(x) \wedge \text{rational}(x))$
- (D) $\exists x (\text{rational}(x) \rightarrow \text{real}(x))$

Explanation

(A) "There exist some numbers which are either real OR rational"

(B) "All real numbers are rational"

(C) "There exist some numbers which are both real AND rational"

(D) "There exist some numbers for which rational implies real"

(See [Propositional Logic](#) for details)

Clearly answer C is correct among all

Practice Question

What is the logical translation of the following statement?

"None of my friends are perfect."

(A) $\exists x(F(x) \wedge \neg P(x))$

(B) $\exists x(\neg F(x) \wedge P(x))$

(C) $\exists x(\neg F(x) \wedge \neg P(x))$

(D) $\neg \exists x(F(x) \wedge P(x))$

☐ A

☐ B

☐ C

☐ D

$F(x) \implies x$ is my friend

$P(x) \implies x$ is perfect

D is the correct answer.

A. There exist some friends which are not perfect

B. There are some people who are not my friend and are perfect

C. There exist some people who are not my friend and are not perfect.

D. There doesn't exist any person who is my friend and perfect

Practice Question

I. $\neg \forall x (P(x))$

II. $\neg \exists x (P(x))$

III. $\neg \exists x (\neg P(x))$

IV. $\exists x (\neg P(x))$

Which of the above two are equivalent?

(A) II and III

(B) I and IV

(C) II and IV

(D) I and III



Thank You

